pendent (or nearly independent) tasks that can then be done in a parallel manner to ensure faster completion. I/O is one such area where partitioning has helped a great deal in ensuring faster communication software for all our mobile devices.

Concurrency

Working on a large number of jobs simultaneously also helps getting more amount of work done faster. Concurrency is of incredible use when doing API calls, or handling user requests (WhatsApp handles two million concurrent text data requests simultaneously and 74 million requests per second).

This, however, can be tricky as when coded incorrectly can very easily corrupt system behaviour as well as persisted data. Locks and monitors should therefore be used as and when necessary but care should be taken to ensure that they are only in use for the least possible time duration.

Extensive automation testing

When you start off with an MVP and are testing everything manually, you can be sure to have release cycles of a reasonable time frame, but once the feature set and the codebase grows over time, manual testing is no longer a viable option. Often at this point, architects design a secondary supporting automation test suite only to realise that their architecture does not support automated tests (at least not effectively). Right from the start, automation should be a design facet for all the components of the application. Each component should ideally have an independent set of test cases that are exhaustive of all of its components. The entire application should have another set of test cases for regression tests.

A well designed automation suite can reduce bugs in the code or even detect if the design of the architecture is compromised.

Conclusion

These simple tips, although evident and obvious, are often ignored or compromised upon for the sake of time constraints or other concerns. Granted these are only guidelines and to follow them to the letter would perhaps increase the design time for an app, sometimes unacceptably so. However, they do help reduce both time and effort in the long run and when followed with enthusiasm they can actually help somebody design a great application that is both scalable and robust. d